Politehnica University of Bucharest

Faculty of Automatic Control and Computer Science

Computer Science Departament

Network monitoring using emerging techniques
Diploma Project - September 2025

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Introduction to Monitoring

Performance

Security

Incident identification and remediation

Reducing downtime

Increase user experience quality

Project Objectives

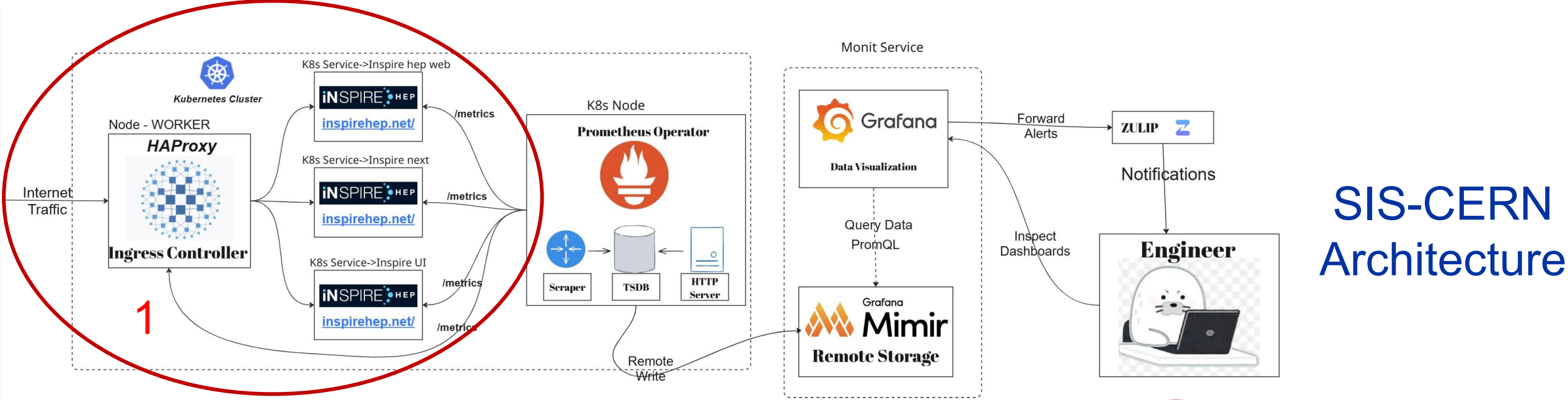
1. Understanding and reproducing the CERN network on UPB systems

2. Creating Dashboards for viewing network components

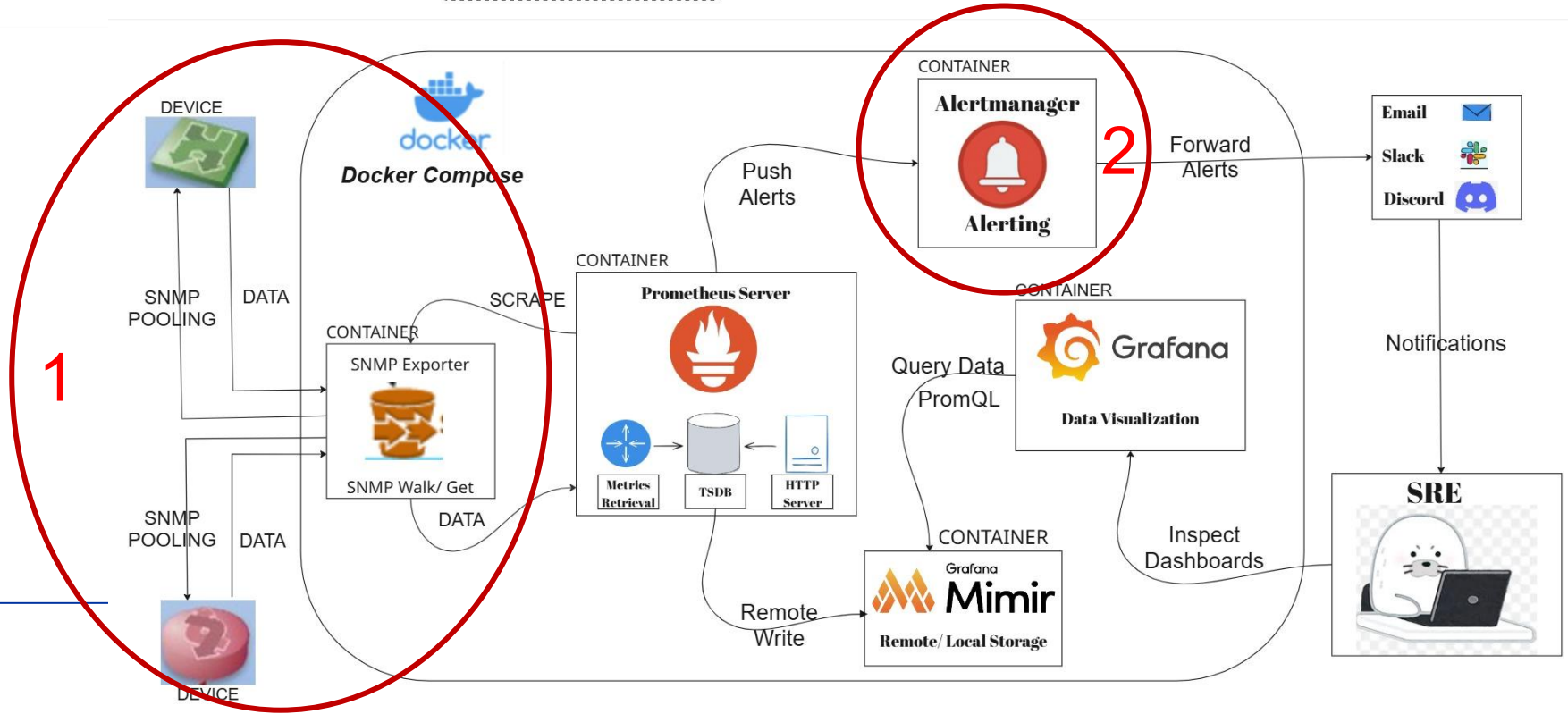
   application status, memory usage, storage space, node status

3. Configure the alert system to generate relevant notifications that detect anomalies early on.

4. Documenting the process and improving the network based on metrics



SIS-CERN Architecture



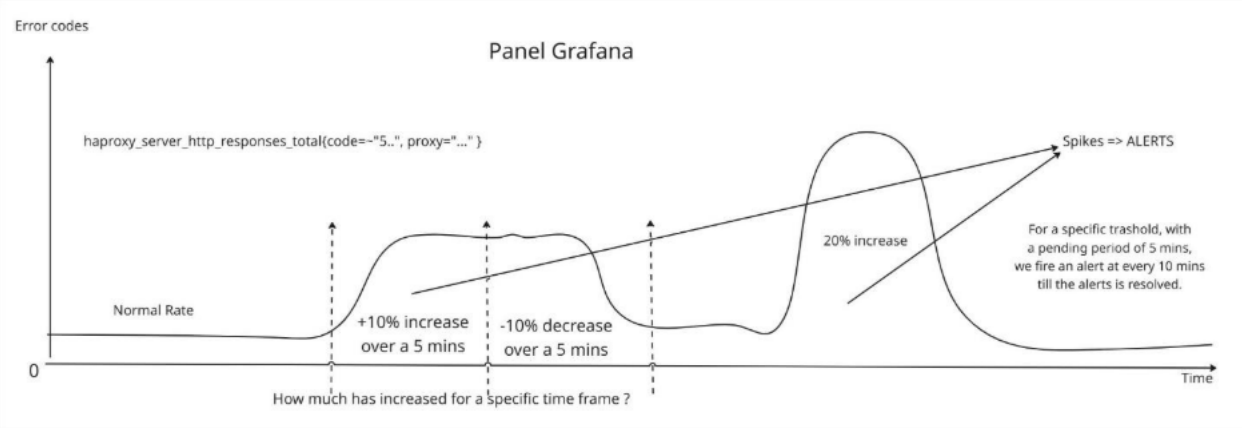
QKD-UPB Architecture

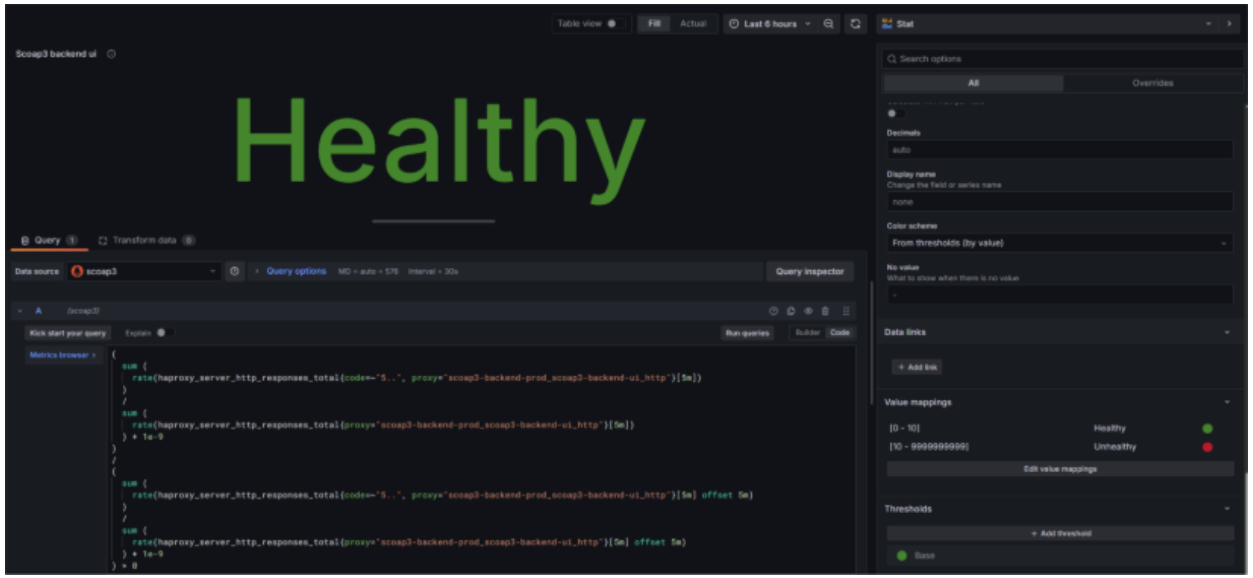
Process

- Infrastructure Analysis
- Identification of data sources (e.g., HAProxy, cAdvisor or Kubelet)
- How to access the data (PromQL queries on TSDB)
- Identifying relevant metrics (perspective)
- How we display and process these ?

HTTP 5xx report per proxy

Measures the increase in 5XX error rate relative to the total number of responses, considering variations in requests and errors over a 5-minute period.

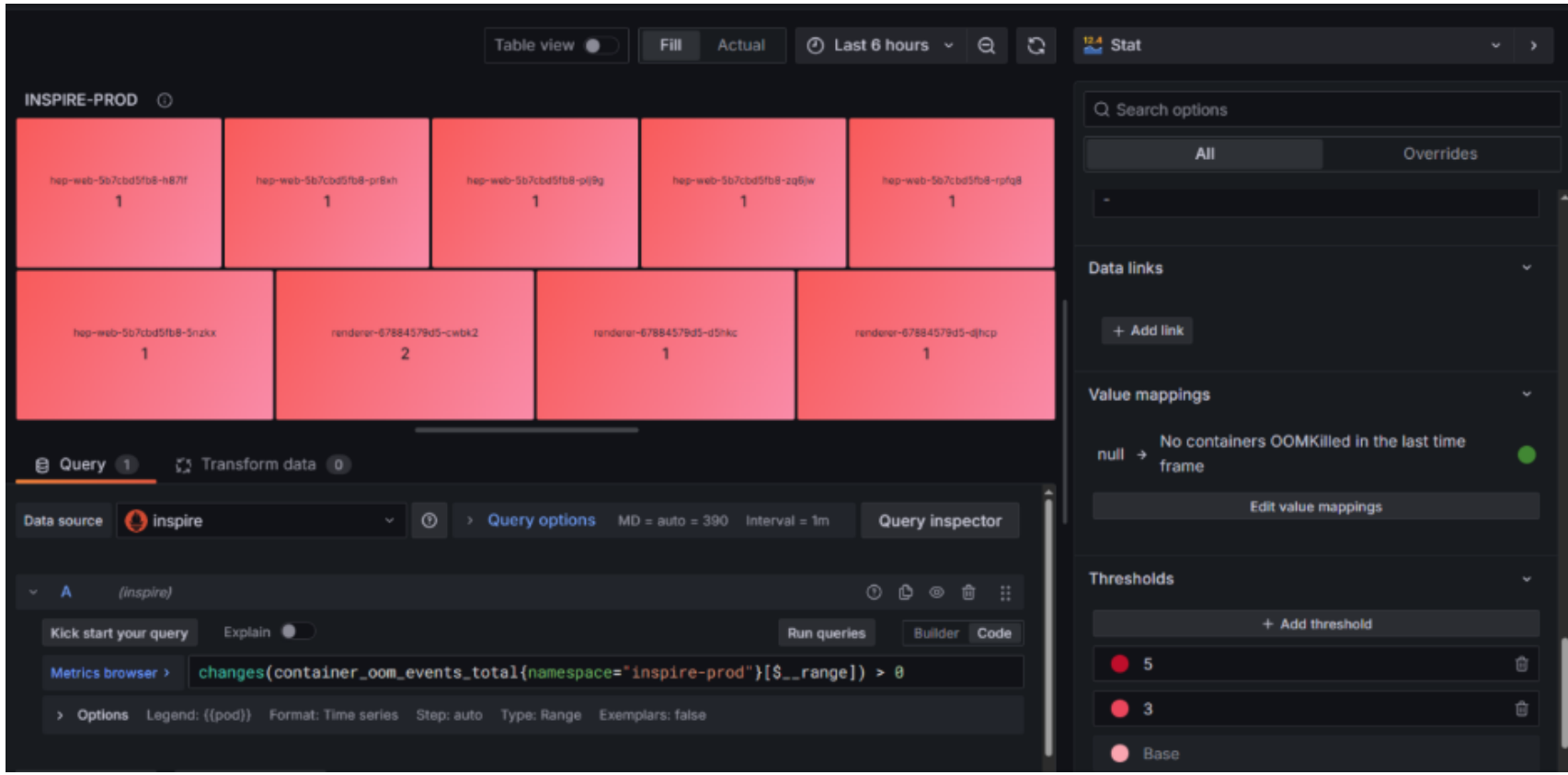




Number of restarts per Pod for OOMKilled events

Monitoring nods restarts caused by OOMKilled events within a specified time interval.

Detects and fixes memory-related problems.



Storage space used

Displays the percentage of total storage space used versus available for each namespace.

For volumes using over 90%, allocate additional space and set up specific alerts.

Alerts

Percentage of total storage used › storage

1 firing | 4 normal | 1m

State	Name	Health	Summary	Next evaluation	Actions
Normal	CAP Total Storage Used	ok		in a few seconds	View Edit More
Normal	HEPData Total Storage Used	ok		in a few seconds	View Edit More
Normal	INSPIRE Total Storage Used	ok		in a few seconds	View Edit More
Normal	OAPEN Total Storage Used	ok		in a few seconds	View Edit More
Firing for 5d 1h 45m	SCOAP3 Total Storage Used	ok		in a few seconds	View Edit More

Show state history

Evaluate: Every 1m
Pending period: 1m
Last evaluation: a few seconds ago
Evaluation time: 9s

Data source: scoap3

Instances: 1 firing | 5 normal

State	Labels	Created
Alerting	persistentvolumeclaim scoap3-harvesting-data +4 common labels	2025-08-29 14:59:00
Normal	persistentvolumeclaim data-scoap3-backend-mq-0 +4 common labels	2025-08-29 14:58:00
Normal	persistentvolumeclaim data-scoap3-backend-redis-0 +4 common labels	2025-08-29 14:58:00
Normal	persistentvolumeclaim data-scoap3-mq-0 +4 common labels	2025-08-29 14:58:00
Normal	persistentvolumeclaim data-scoap3-redis-0 +4 common labels	2025-08-29 14:58:00
Normal	persistentvolumeclaim scoap3-data +4 common labels	2025-08-29 14:58:00

Alerts in Grafana
+ Alertmanager

Notifications on
Discord and Zulip

[FIRING:1] status-update (172.20.250.12 snmp-exporter-cerberis3 cerberis3_qkd_node critical)

Alerts Firing:
Labels:
- alertname = status-update
- instance = 172.20.250.12
- job = snmp-exporter-cerberis3
- module = cerberis3_qkd_node
- severity = critical

Annotations:
- description = Endpoint down
- title = Status update for 172.20.250.12

Source: http://qkd-idq-qms:9090/graph?g0.expr=(up%7Bmodule%3D"cerberis3_qkd_node"%7D+<+1)&g0.tab=1

[RESOLVED] status-update (172.20.250.12 snmp-exporter-cerberis3 cerberis3_qkd_node critical)

Alerts Resolved:
Labels:
- alertname = status-update
- instance = 172.20.250.12
- job = snmp-exporter-cerberis3
- module = cerberis3_qkd_node
- severity = critical

Annotations:
- description = Endpoint down
- title = Status update for 172.20.250.12

Source: http://qkd-idq-qms:9090/graph?g0.expr=(up%7Bmodule%3D"cerberis3_qkd_node"%7D+<+1)&g0.tab=1

Results

The system provides real-time visibility into infrastructure.

Improved response time through automated alerts and early incident prevention.

All components are monitored and integrated into the network, and metrics are collected in real time.

Further Developments

- Transforming metrics in Prometheus to improve alerts
- Infrastructure consolidation by eliminating SPOF
- Integrating AI for metric analysis and obtaining broader results

Dashboards

Alertmanager

Q & A

Grafana

Containerization

Kubernetes

Mimir

Thank you!

Prometheus

Monitoring